DATA MINING COURSE PROJECT

Voting-Pattern Analysis of Knesset Elections

Exploratory data analysis and principal component analysis of the 24th and 25th Knesset election results

Python · pandas · scikit-learn · seaborn · matplotlib

OVERVIEW

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Source tables

Four tables describe the two elections, at ballot-box and locality level. Sizes exclude row 1, which holds the field names.

Table	Dataset size	Each row represents
Results by ballot box, Knesset 24	646,300 (12,926 records × 50 variables)	One physical ballot box (“barzel”) within a polling committee; a committee may include several ballot boxes
Results by locality, Knesset 24	55,890 (1,215 records × 46 variables)	One locality, with count variables describing its voting results by category (valid and invalid votes, and votes per party) in a given election
Results by ballot box, Knesset 25	639,795 (12,545 records × 51 variables)	One physical ballot box within a polling committee
Results by locality, Knesset 25	57,152 (1,216 records × 47 variables)	One locality, with the same count variables

Q2(A) I–III

Variables: ballot-box tables

Variable	Meaning
Committee code	Administrative identifier of the polling committee; not unique at national level
Barzel	Identifier of a physical ballot box within the polling committee; one committee may include several
Locality name / code	Name of the locality where the polling committee operates, and its official numeric code
Kalpi	Internal identifier of the polling station within the locality
Rikuz	Organizational variable used to group polling stations for management and counting
Shofet	Votes received in judges' envelopes
BZB	Votes received in BZB envelopes (special polling stations, such as for soldiers, hospital patients and prisoners)
Voters / Invalid / Valid	Total voters in the ballot box, invalid votes, and valid votes
Party columns	Each column is one party; each cell is the number of valid votes for that party in that ballot box

The Knesset 25 table uses the same variables, except that several parties changed relative to Knesset 24.

Q2(A) I–III

Variables: locality tables

Variable	Meaning
Committee code	Identifier of the election committee
Locality name	Name of the city or council
Locality code	Unique numeric identifier of the locality
BZB	Number of eligible voters in the locality
Voters	Number of people who actually voted in the locality
Invalid	Number of invalid votes in the locality
Valid	Number of valid votes in the locality
Party columns	Number of votes each party received in the locality; together they describe the distribution of party voting

Q2(B) IV

Ballot-box dataset, Knesset 25: structure

Dataset size	Variable type	Missing values
152,191 (11,707 records × 13 variables, excluding the field-name row)	int64 (all party columns)	0

The cleaned dataset holds the vote counts of 13 parties in each ballot box:

- Labor
- Jewish Home
- United Torah Judaism
- Balad
- Hadash-Ta'al
- Religious Zionism
- National Unity
- Yisrael Beiteinu
- Likud
- Meretz
- United Arab List (Ra'am)
- Yesh Atid
- Shas

Q2(B) V

Ballot-box dataset, Knesset 25: descriptive statistics

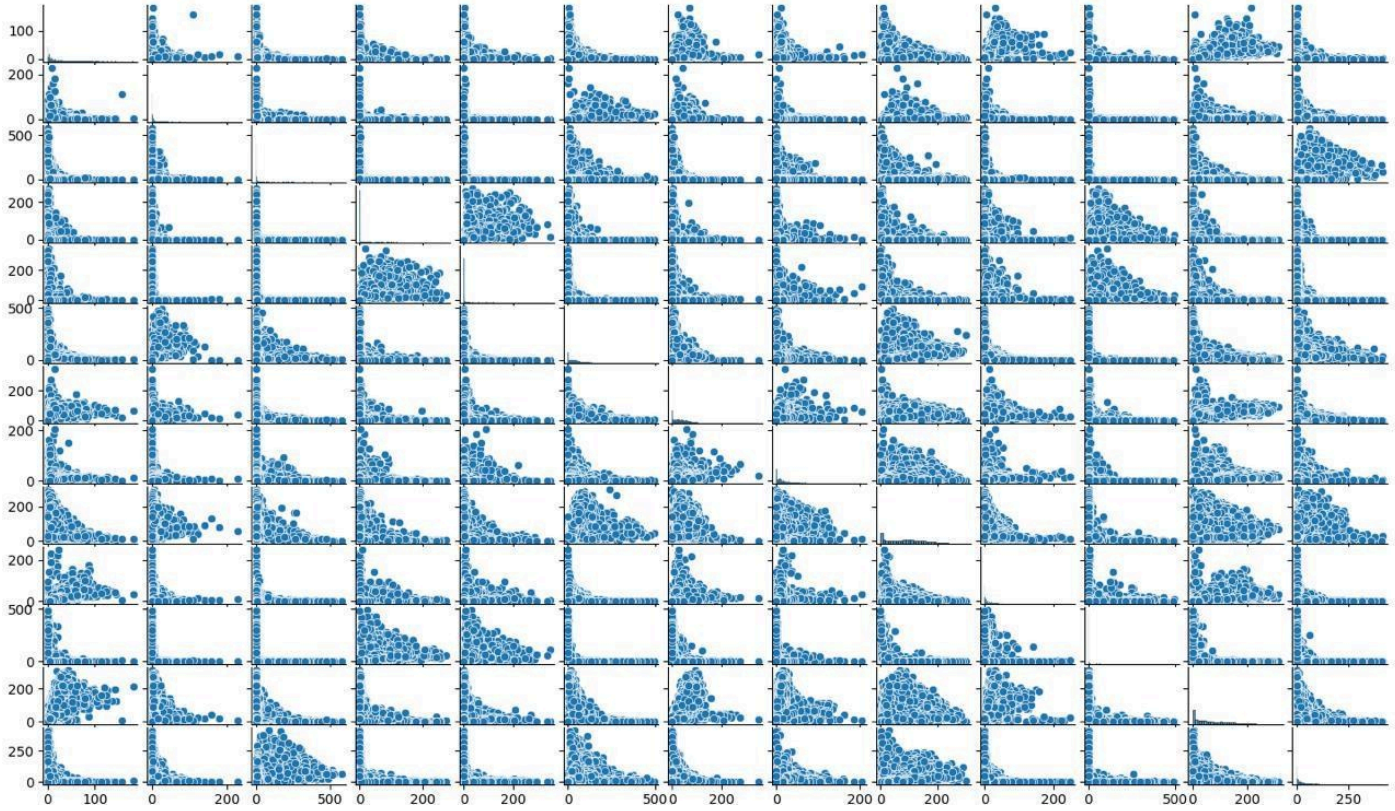
The `describe()` function provides basic statistics for numeric variables, in this case the votes each party received per ballot box.

	Meretz	Ra'am	Yesh Atid	Shas
count	11,707	11,707	11,707	11,707
mean	11.65	15.89	65.75	30.89
std	17.44	48.27	61.14	44.32
min	0	0	0	0
25%	1	0	12	3
50%	5	0	49	15
75%	15	1	108	39
max	250	494	313	417

Statistic	Meaning	Yesh Atid	Meretz
count	Available observations	11,707; complete in all records	11,707; no missing values
mean	Arithmetic mean of votes	≈ 65.75 per ballot box; high average concentration	≈ 11.65; much lower than Yesh Atid
std	Dispersion around the mean	≈ 61.14; high variance between ballot boxes	≈ 17.44; more moderate spread
min	Lowest observed value	0; some ballot boxes have no votes for the party	0; here too
25%	First quartile	12; a quarter of ballot boxes have up to 12 votes	1; a quarter have up to one vote
50%	Median	49; half of ballot boxes have up to 49 votes	5; half have up to 5 votes
75%	Third quartile	108; most ballot boxes show high concentration	15; relatively low spread
max	Highest observed value	313; especially high concentration	250; isolated extreme values

Q2(B) V

Scatter matrix, Knesset 25



Negative relationship: Likud and Yesh Atid

The scatter plots show a clear downward slope. Ballot boxes with many votes for Likud show few for Yesh Atid, and vice versa. The two parties address opposing ideological audiences, and the relationship reflects direct competition for leadership of the political blocs.

Positive relationship: Shas and United Torah Judaism

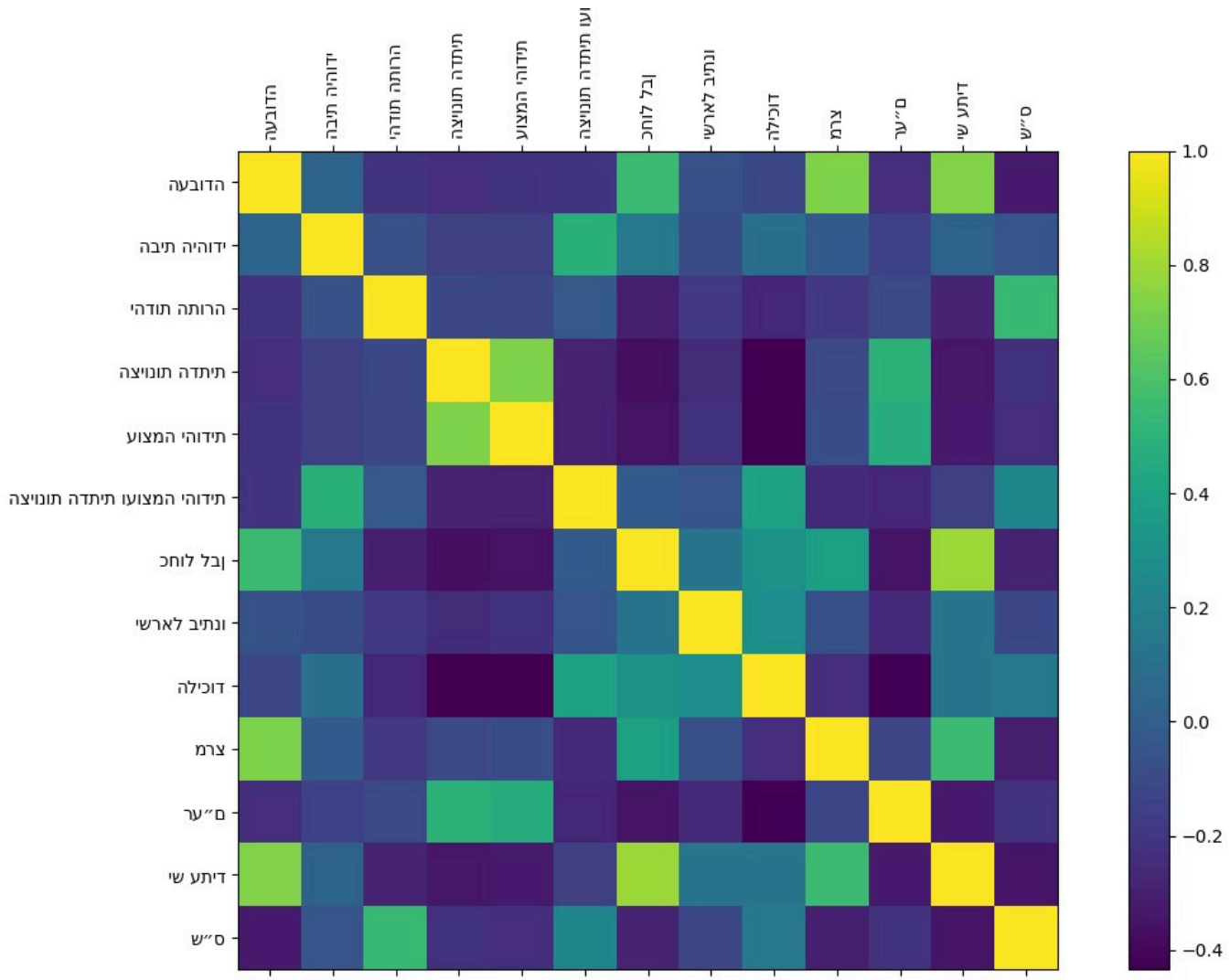
Ballot boxes with high support for Shas also tend to show high support for United Torah Judaism. Both parties address a religious, Haredi public, so their voting patterns are similar.

Figure labels, Knesset 25

הליכוד	Likud	יש עתיד	Yesh Atid
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	רע"ם	United Arab List (Ra'am)
הבית היהודי	Jewish Home	כחול לבן	National Unity (code כן)
עוצמה יהודית	Hadash-Ta'al (code ום)	הציונות הדתית	Balad (code ד)
הציונות הדתית ועוצמה יהודית	Religious Zionism (code ט)		

Q2(B) V

Correlation matrix, Knesset 25



The matrix shows the strength and direction of the linear relationship between party vote counts.

Positive correlation: Yesh Atid and National Unity

A light color indicates positive correlation. Where Yesh Atid receives many votes, National Unity tends to receive more as well, indicating a similar voting pattern at ballot-box level.

Negative correlation: Likud and Meretz

A dark color indicates negative correlation. Where Likud is strong, Meretz tends to be weak, and vice versa, reflecting political polarization between the two electorates.

Figure labels, Knesset 25

הליכוד	Likud	יש עתיד	Yesh Atid
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	רע"ם	United Arab List (Ra'am)
הבית היהודי	Jewish Home	כחול לבן	National Unity (כן)
עוצמה יהודית	Hadash-Ta'al (חם)	הציונות הדתית	Balad (ד)
הציונות הדתית ועוצמה יהודית	Religious Zionism (ט)		

Q2(B) VI

Ballot-box dataset, Knesset 24: structure

Dataset size	Variable type	Missing values
157,651 (12,127 records × 13 variables, excluding the field-name row)	int64 (all party columns)	0

The cleaned dataset holds the vote counts of 13 parties in each ballot box:

Labor · Yamina · United Torah Judaism · Joint List · Religious Zionism · Blue and White · Yisrael Beiteinu · Likud · Meretz · United Arab List (Ra'am) · Yesh Atid · Shas · New Hope

Q2(B) VI

Ballot-box dataset, Knesset 24: descriptive statistics

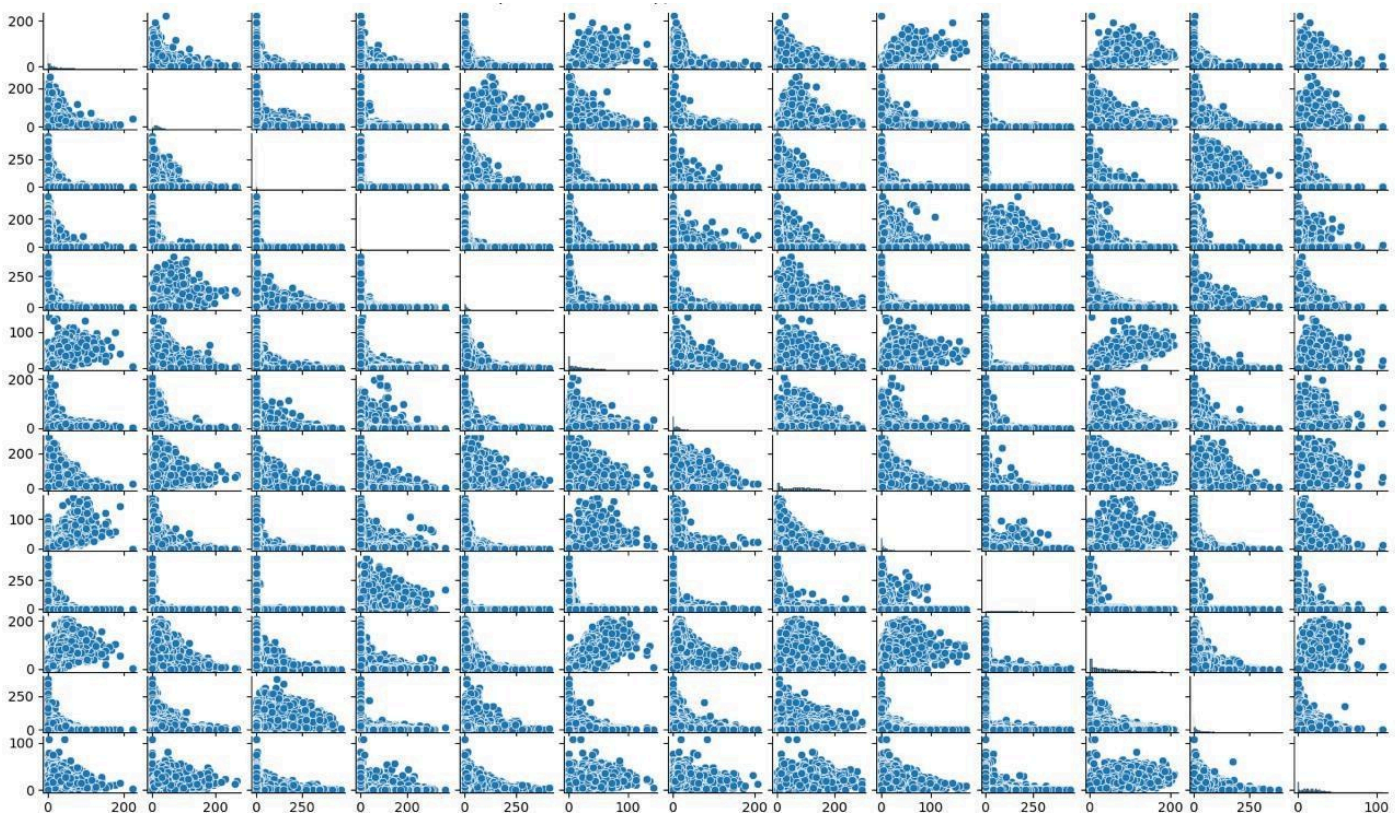
The `describe()` function provides basic statistics for numeric variables, in this case the votes each party received per ballot box.

	Labor	Yamina	…	Shas	New Hope
count	12,127	12,127	…	12,127	12,127
mean	19.34	19.31	…	24.08	15.43
std	22.80	21.80	…	35.60	11.62
min	0	0	…	0	0
25%	3	6	…	2	5
50%	10	15	…	12	15
75%	29	26	…	30	23
max	223	265	…	381	108

Statistic	Meaning	Labor	Shas
count	Available observations	12,127; complete in all records	12,127; no missing values
mean	Arithmetic mean of votes	≈ 19.34 per ballot box	≈ 24.08; higher than Labor
std	Dispersion around the mean	≈ 22.80; moderate spread	≈ 35.60; higher by comparison
min	Lowest observed value	0; some ballot boxes have no votes for the party	0; here too
25%	First quartile	3; a quarter of ballot boxes have up to 3 votes	2; a quarter have up to two votes
50%	Median	10; half have up to 10 votes	12; half have up to 12 votes
75%	Third quartile	29	30
max	Highest observed value	223	381; almost twice as high

Q2(B) VI

Scatter matrix, Knesset 24



Strong negative relationship: competition between parties

Plots of parties from the same bloc sometimes show a descending cloud. Where one party is strong the other is weak, meaning the parties compete for the same voters and one gains at the other's expense.

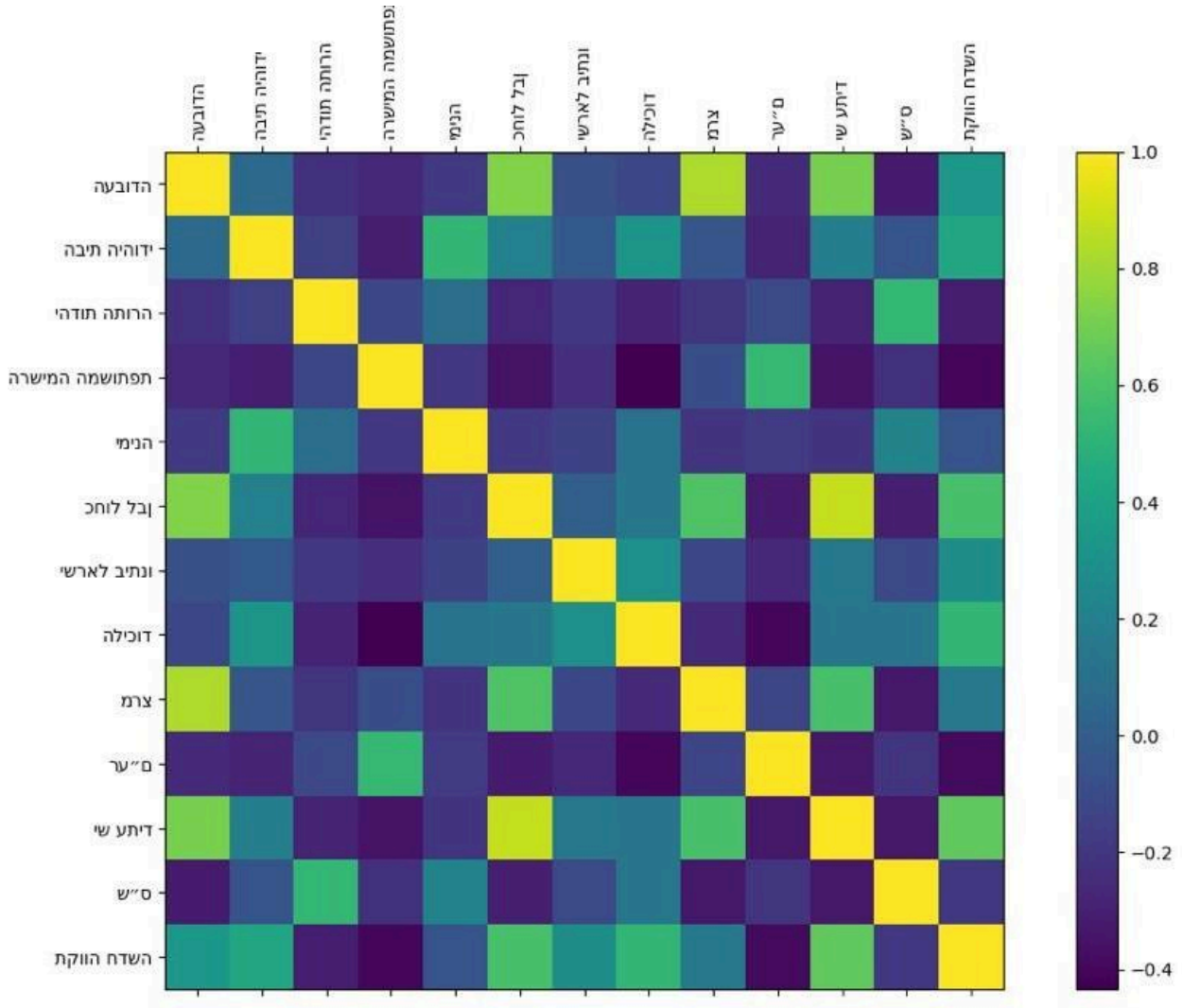
Points concentrated near the axes: sectoral parties

For sectoral parties, most points are close to zero with a few high values. Most ballot boxes barely vote for the party, while in certain ballot boxes it is very strong, which is typical of a focused audience.

Figure labels, Knesset 24

הליכוד	Likud	יש עתיד	Yesh Atid
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	רע"ם	United Arab List (Ra'am)
כחול לבן	Blue and White	תקווה חדשה	New Hope
הרשימה המשותפת	Joint List	הבית היהודי	Yamina (code ב)
ימינה	Religious Zionism (code ט)		

Correlation matrix, Knesset 24



Positive correlation: Yesh Atid and Blue and White

A light color indicates positive correlation. Where Yesh Atid receives many votes, Blue and White also tends to, suggesting that the two parties address audiences with shared ideological or socio-demographic characteristics.

Negative correlation: Likud and Meretz

A dark color indicates negative correlation. Where Likud is strong, Meretz tends to be weak, and vice versa. This reflects clear polarization and ideological competition for different audiences at ballot-box level.

Figure labels, Knesset 24

הליכוד	Likud	יש עתיד	Yesh Atid
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	רע"ם	United Arab List (Ra'am)
כחול לבן	Blue and White	תקווה חדשה	New Hope
הרשימה המשותפת	Joint List	הבית היהודי	Yamina (code ב)
ימינה	Religious Zionism (code ט)		

Q2(B) VI

Comparing the 2024 and 2025 datasets

Similarities

1. **Similar polarization.** Both years show clear polarization between right-wing and left/center blocs.
2. **Stable sectoral patterns.** Sectoral parties retain concentrated support in specific ballot boxes.
3. **Relationships between close parties.** Parties with similar target audiences are positively correlated.

Differences

1. **Change in party composition.** Mergers and splits in Knesset 25 changed the distribution of votes within the blocs.
2. **Changes in large localities.** Large localities showed more noticeable shifts in voting patterns.
3. **Change in strength of relationships.** Some correlations strengthened or weakened; the 2024 matrix is more colorful.

Q3(A)

Organizing the data to compare parties

Hypothesis: similarity between parties is expressed in similar voting patterns among similar voter populations.

The original table is localities × parties, with one locality per row and one party per column. To compare parties, the table is transposed so that each party becomes the unit of observation.

- **Observations:** parties, one per row
- **Features:** localities or ballot boxes, one per column

Each party is then represented by a vector of voting results across all localities or ballot boxes, and the comparison is made on this representation.

Before (localities × parties)

	A	B	C
X	120	80	20
Y	60	140	10

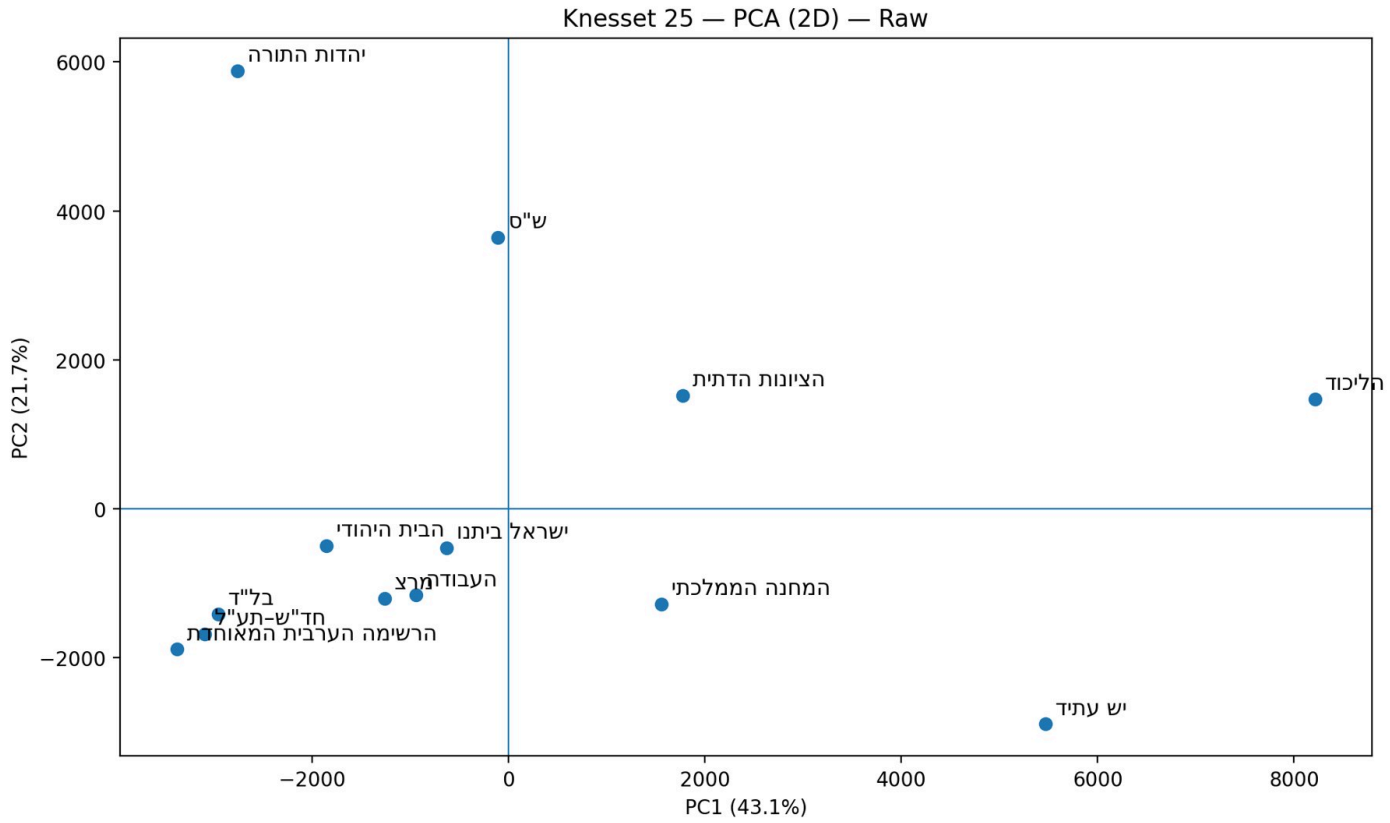
After transpose (parties × localities)

	X	Y
A	120	60
B	80	140
C	20	10

Parties become the objects of comparison, and localities become the features that describe them.

Q3(B) 1–2

Two-dimensional PCA, no scaling



PC1	PC2	Combined
43.1%	21.7%	64.9%

The map identifies groups of parties with similar voting patterns (close together) and those with different patterns (far apart). **PC1** separates ballot boxes where Arab parties stand out from those where Jewish-Zionist parties stand out. **PC2** contrasts religious and Haredi patterns with secular and Arab patterns.

PC1 extremes

Party	Score
United Arab List	−3,381.69
Hadash-Ta'al	−3,100.36
Balad	−2,962.46
Religious Zionism	1,772.38
Yesh Atid	5,471.99
Likud	8,213.47

PC2 extremes

Party	Score
Yesh Atid	−2,882.34
United Arab List	−1,879.52
Hadash-Ta'al	−1,683.61
Religious Zionism	1,523.98
Shas	3,640.87
United Torah Judaism	5,884.68

Q3(B) 3

Min-Max scaling: method

Min-Max scaling is applied to each ballot box (column) *j*:

$$X'_{ij} = (X_{ij} - min_j) / (max_j - min_j)$$

- The smallest value becomes 0
- The largest value becomes 1
- All others fall between 0 and 1

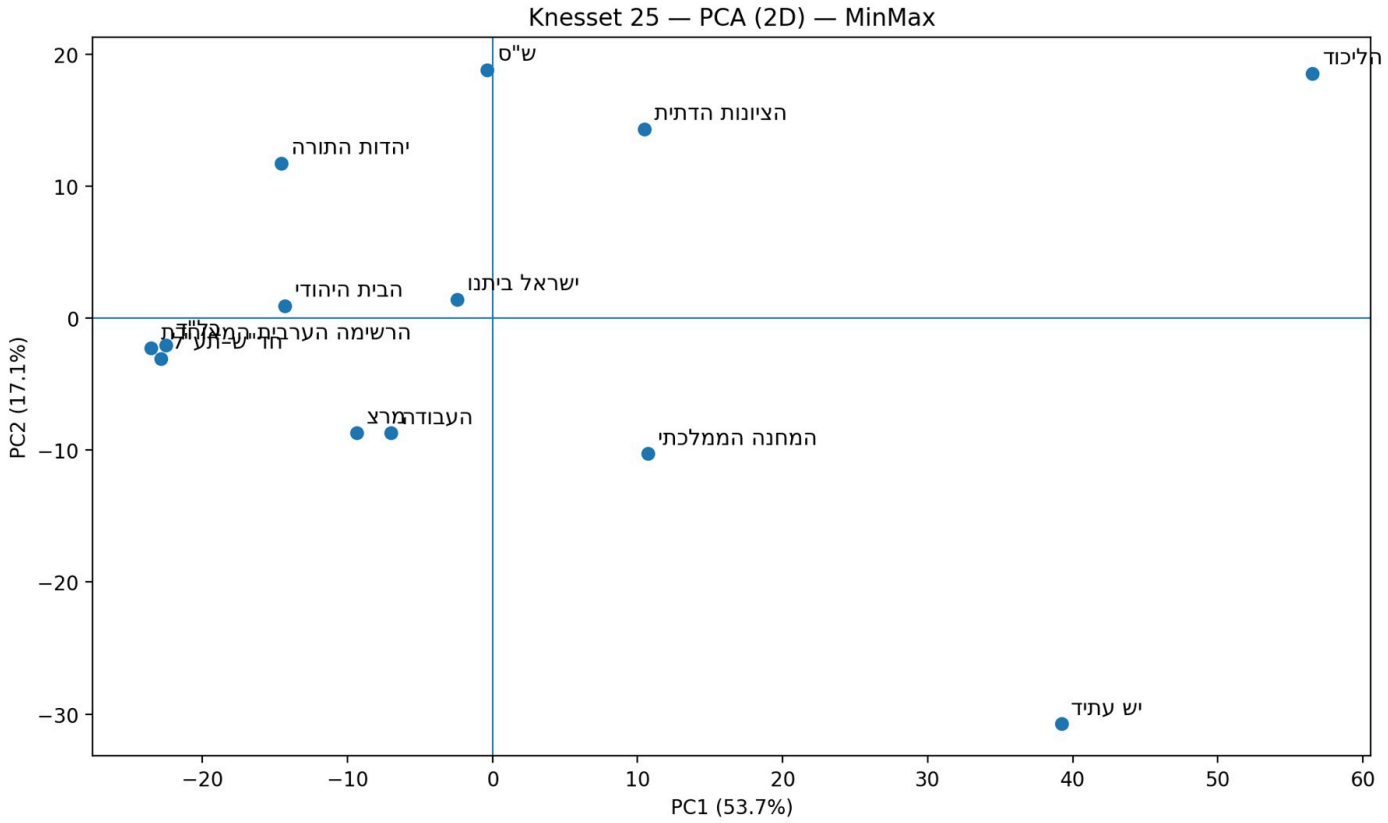
Why it matters

Scaling neutralizes ballot-box size (large against small) and highlights the relative pattern of votes within each ballot box.

PCA now compares parties by relative profile, rather than by who receives the most votes because of having many voters.

Q3(B) 3

Two-dimensional PCA, Min-Max scaling



PC1	PC2	Combined
53.7%	17.1%	70.8%

PC1 still separates Arab parties from central Jewish parties: the further right a point, the more “Jewish-Zionist” the pattern. **PC2** sharpens into a center-left and right/religious-traditional axis. The comparison is now based on relative deviations within each ballot box rather than absolute counts.

PC1 extremes

Party	Score
United Arab List	−23.55
Hadash-Ta'al	−22.83
Balad	−22.53
National Unity	10.72
Yesh Atid	39.21
Likud	56.53

PC2 extremes

Party	Score
Yesh Atid	−30.70
National Unity	−10.28
Labor	−8.68
Religious Zionism	14.32
Likud	18.53
Shas	18.82

Z-score standardization: method

For each ballot box *j*, the mean and standard deviation across parties are calculated, and each value is converted to a Z-score:

$$Z_{ij} = (X_{ij} - \mu_j) / \sigma_j$$

- **Positive:** the party received more than that ballot box's average
- **Negative:** the party received less than the average
- **Around ±2:** an outlier, far above or below what is usual in that ballot box
- **Close to 0:** about average

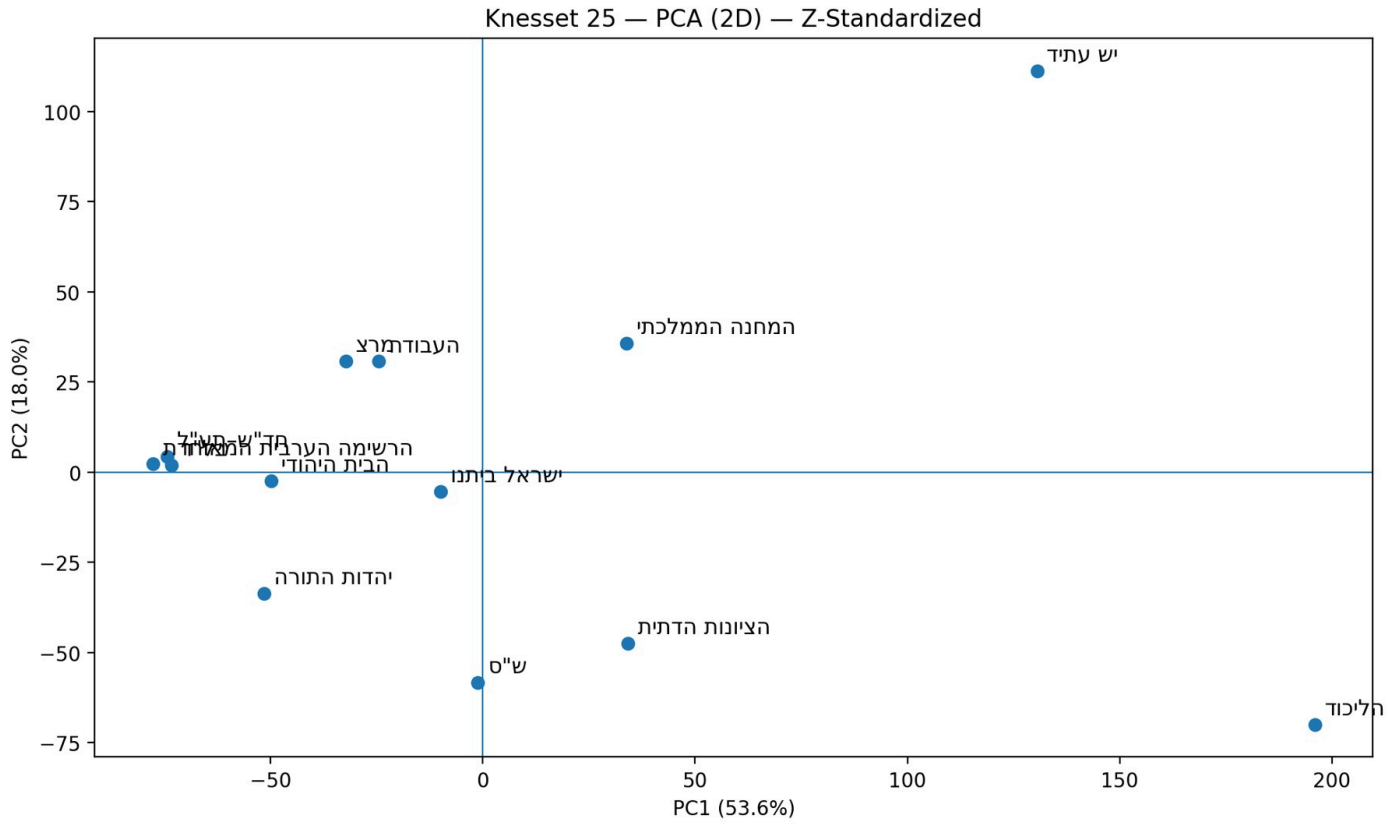
Why it matters

Standardization removes the effect of ballot-box size (300 voters against 3,000) and highlights the relative voting profile of each ballot box: who is dominant and who is weak.

PCA then finds axes that separate parties according to these relative patterns.

Q3(B) 4

Two-dimensional PCA, Z-score standardization



PC1	PC2	Combined
53.6%	18.0%	71.6%

PCA is now based on standard scores, so it emphasizes relative patterns rather than vote counts or ballot-box size. **PC1** mainly separates Arab parties from mainstream Jewish parties. **PC2** mainly separates the right/religious-traditional pattern (low) from the center-left pattern (high).

PC1 extremes

Party	Score
United Arab List	−77.69
Hadash-Ta'al	−74.32
Balad	−73.32
Religious Zionism	34.18
Yesh Atid	130.50
Likud	195.90

PC2 extremes

Party	Score
Likud	−69.93
Shas	−58.42
Religious Zionism	−47.50
Meretz	30.87
National Unity	35.78
Yesh Atid	111.32

Q3(B) 5

Preferred method: Z-score standardization

Answers the question

After transposing, each variable is a ballot box, and a party's Z value shows how far it is above or below that ballot box's mean in standard deviations. The comparison is of relative voting patterns, not vote totals.

Best 2D representation

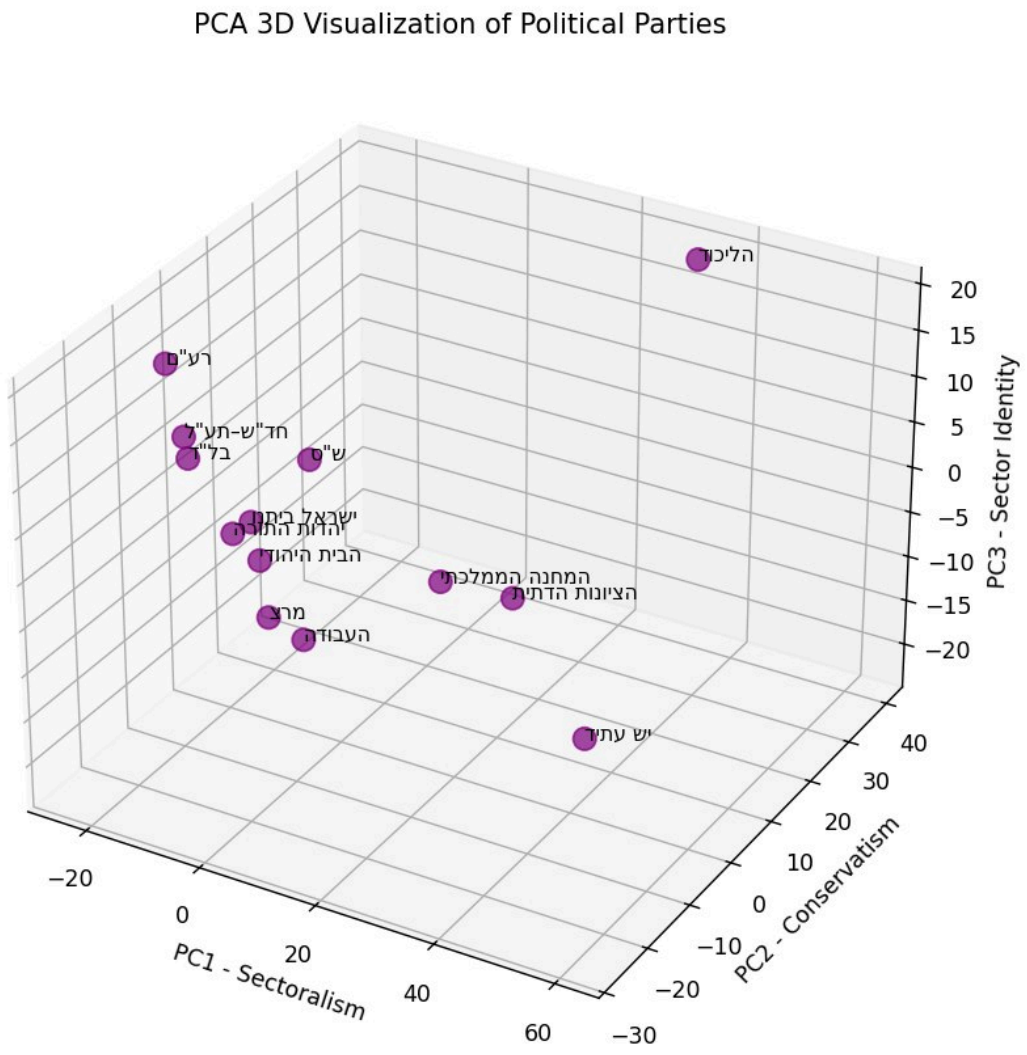
The first two components together explain about 71.6% of the variance, the highest of the three methods, so the two-dimensional representation is of higher quality.

Interpretable axes

The axes still explain what separates groups of parties, such as Arab against Jewish or right-traditional against center-left. Similarity is easy to read: parties close together have similar voting profiles.

Q3(B) 6

Three-dimensional PCA of parties (Z-score)



PC1: sectoral axis

Separates mainstream parties with broad national support, such as Likud and Yesh Atid, from sectoral parties that draw most of their strength from homogeneous, closed localities (Haredi and Arab).

PC2: conservative against liberal axis

Separates, within the Jewish-Zionist sector, the secular-liberal bloc (Meretz, Labor, Yesh Atid) from the religious-conservative bloc (Religious Zionism, Shas, United Torah Judaism).

PC3: sector identity axis

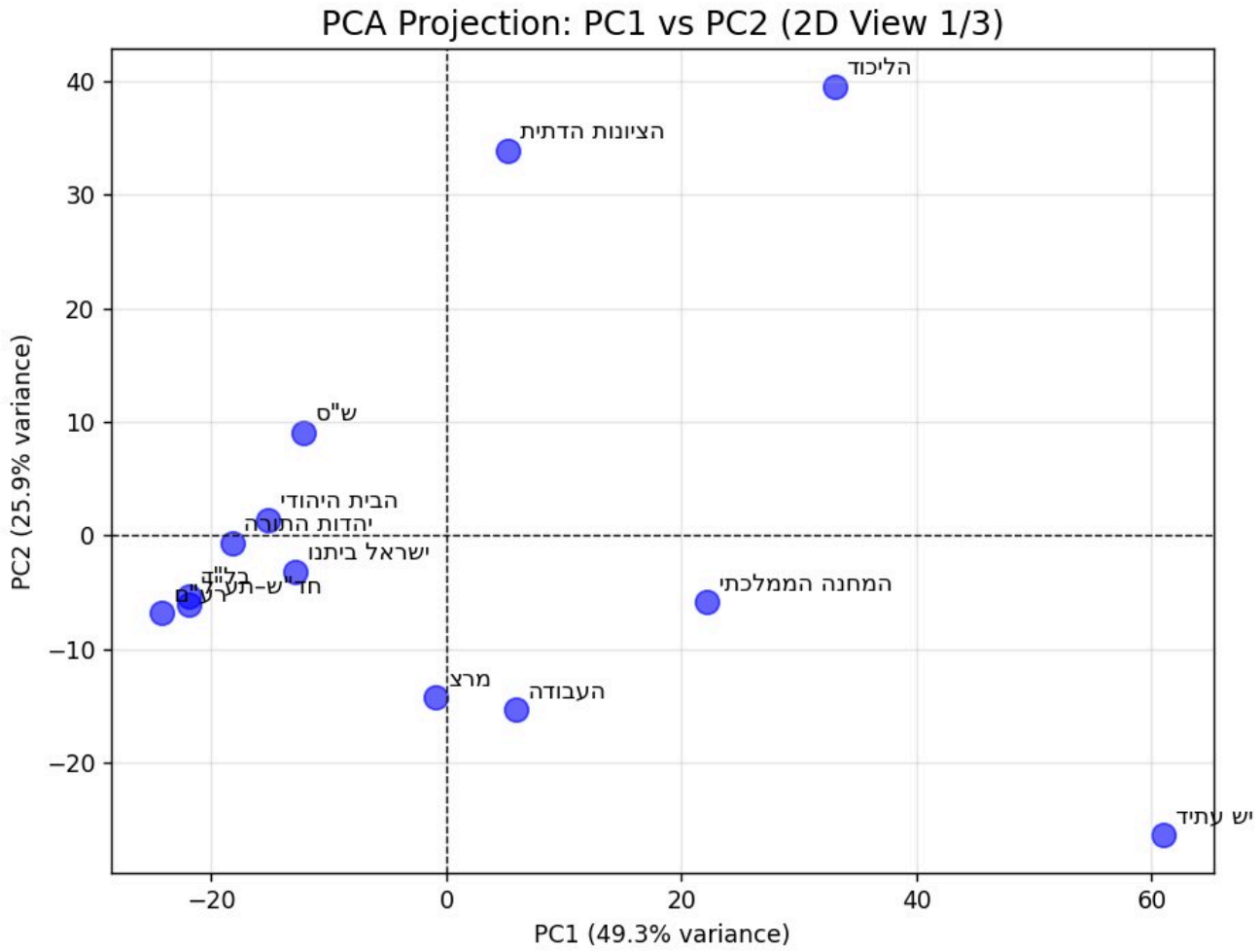
Distinguishes within the sectoral groups between Arab-sector (Ra'am, Hadash-Ta'al) and Haredi-sector (United Torah Judaism, Shas) voting patterns.

Figure labels

הליכוד	Likud	יש עתיד	Yesh Atid
הציונות הדתית	Religious Zionism	המחנה הממלכתי	National Unity
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	חד"ש-תע"ל	Hadash-Ta'al
הרשימה הערבית המאוחדת	United Arab List (Ra'am)	בל"ד	Balad
הבית היהודי	Jewish Home		

Q3(B) 6

Party PCA: PC1 and PC2



Identity against ideology

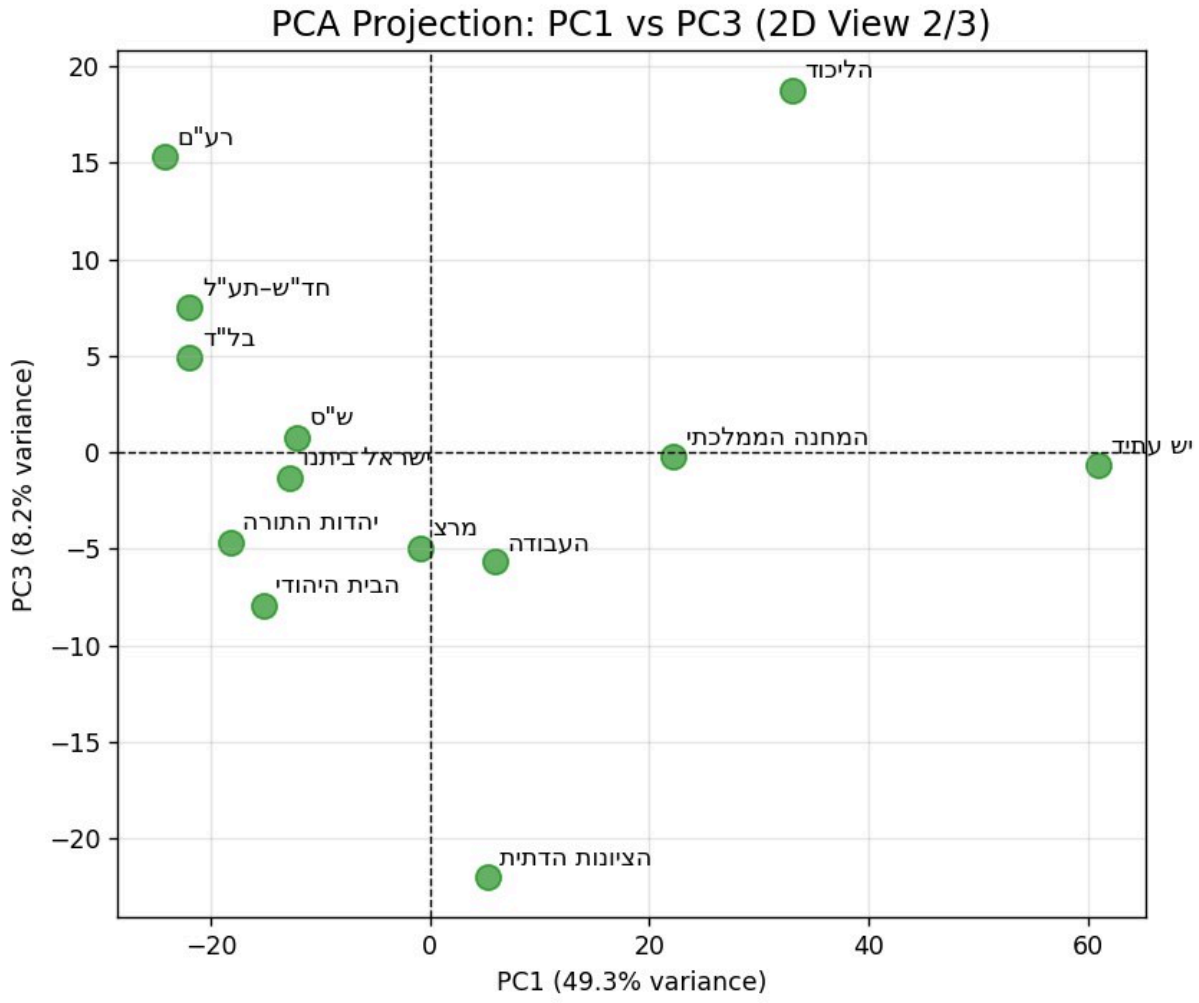
PC1 determines whether a party is sectoral and isolated or nationwide, while PC2 determines whether its values are religious-conservative or secular-liberal.

Together they separate parties that are similar in voter structure but far apart in worldview.

Figure labels

הליכוד	Likud	יש עתיד	Yesh Atid
הציונות הדתית	Religious Zionism	המחנה הממלכתי	National Unity
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	חד"ש-תע"ל	Hadash-Ta'al
הרשימה הערבית המאוחדת	United Arab List (Ra'am)	בל"ד	Balad
הבית היהודי	Jewish Home		

Party PCA: PC1 and PC3



Sectoral strength against sector identity

PC1 separates general from sectoral parties, while PC3 distinguishes within the sectors between Arab society and Haredi society.

Together they show that although both sectors share a homogeneous locality voting pattern (PC1), they are entirely distant in national and community identity (PC3).

Figure labels

הליכוד	Likud	יש עתיד	Yesh Atid
הציונות הדתית	Religious Zionism	המחנה הממלכתי	National Unity
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	חד"ש-תע"ל	Hadash-Ta'al
הרשימה הערבית המאוחדת	United Arab List (Ra'am)	בל"ד	Balad
הבית היהודי	Jewish Home		

Party PCA: PC2 and PC3, and groups of similar parties

PC2 and PC3: culture and identity

PC2 separates worldviews (secular-liberal against religious-conservative), while PC3 performs the final national-sectoral separation.

Together they distinguish between similar conservative parties belonging to entirely different sectors, such as the Haredi and Arab parties.

Can groups of similar parties be identified?

Yes, clearly. PCA translates the sociological reality described by Rivlin into mathematical distances. Parties cluster according to the education and residential systems of their voters:

- **Haredi and Arab parties:** isolated because of locality homogeneity
- **Religious Zionism:** positioned as a conservative but integrated factor
- **General public (secular and traditional):** concentrated at the center of the map because of wide geographic distribution

Q3(C) I

Three-dimensional PCA of ballot boxes, 2025

PC1

Center-liberal against Arab sector

The most significant axis. The positive pole holds ballot boxes with high voting for center and Zionist-left parties (Yesh Atid, National Unity, Labor, Meretz); the negative pole holds Arab-sector ballot boxes (Balad, Hadash-Ta'al, Ra'am).

PC2

Religious-right against left and Arab

Reflects the deep ideological division in Israeli society. The negative pole is identified with the right-wing and Haredi bloc (Shas, Religious Zionism, Likud, United Torah Judaism); the positive pole with left-wing and Arab parties.

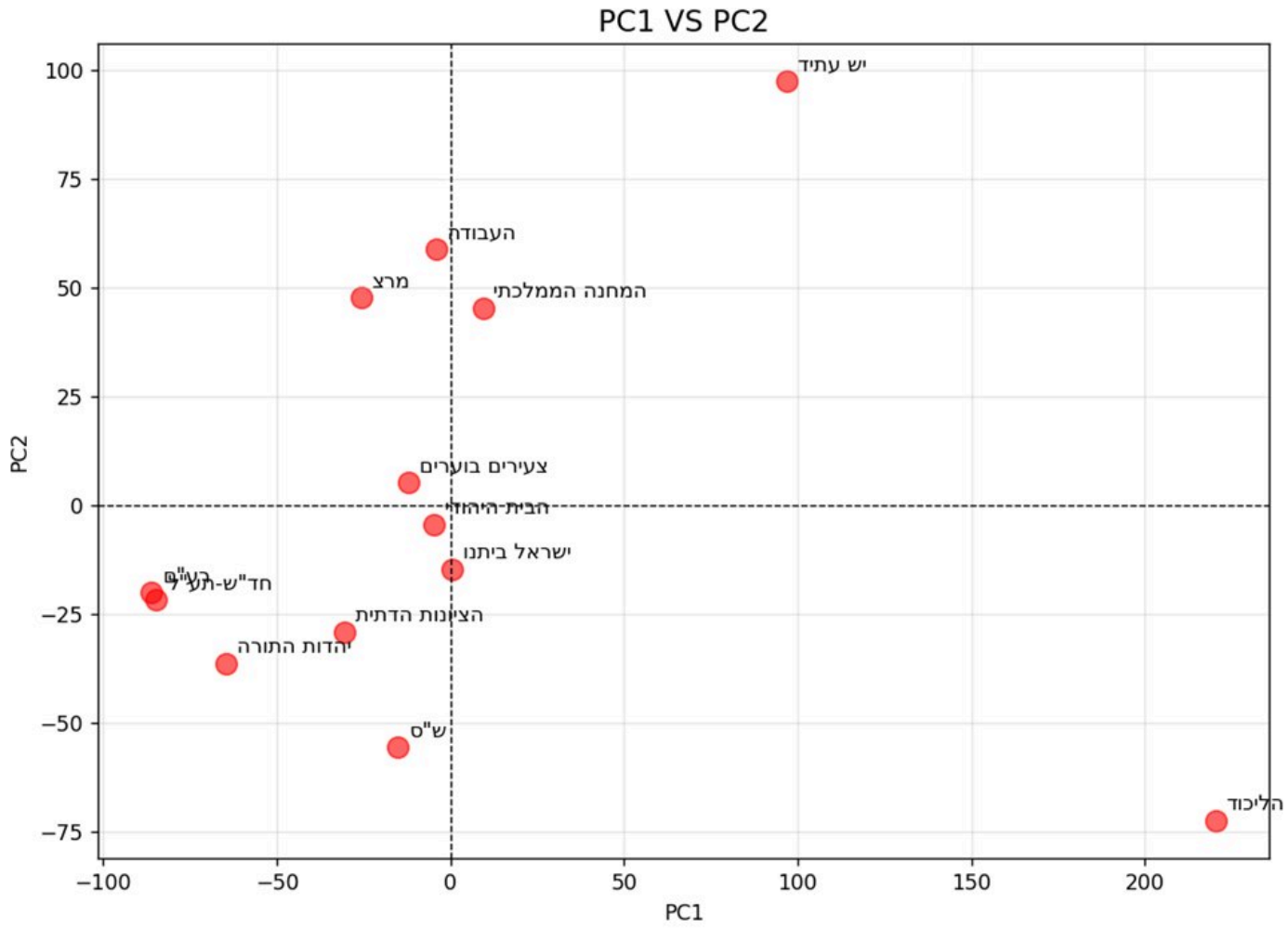
PC3

Identity within the right

Once the large clusters are separated, PC3 distinguishes within the religious right. The negative pole holds the Haredi parties; the positive pole holds the secular and national-religious right (Likud, Yisrael Beiteinu, Religious Zionism).

Q3(C)

Ballot-box PCA: PC1 and PC2



The chart shows the interaction between the national and ideological divides. Together they create a triangular structure that clearly separates three main centers of power: religious right, center-left and Arab parties.

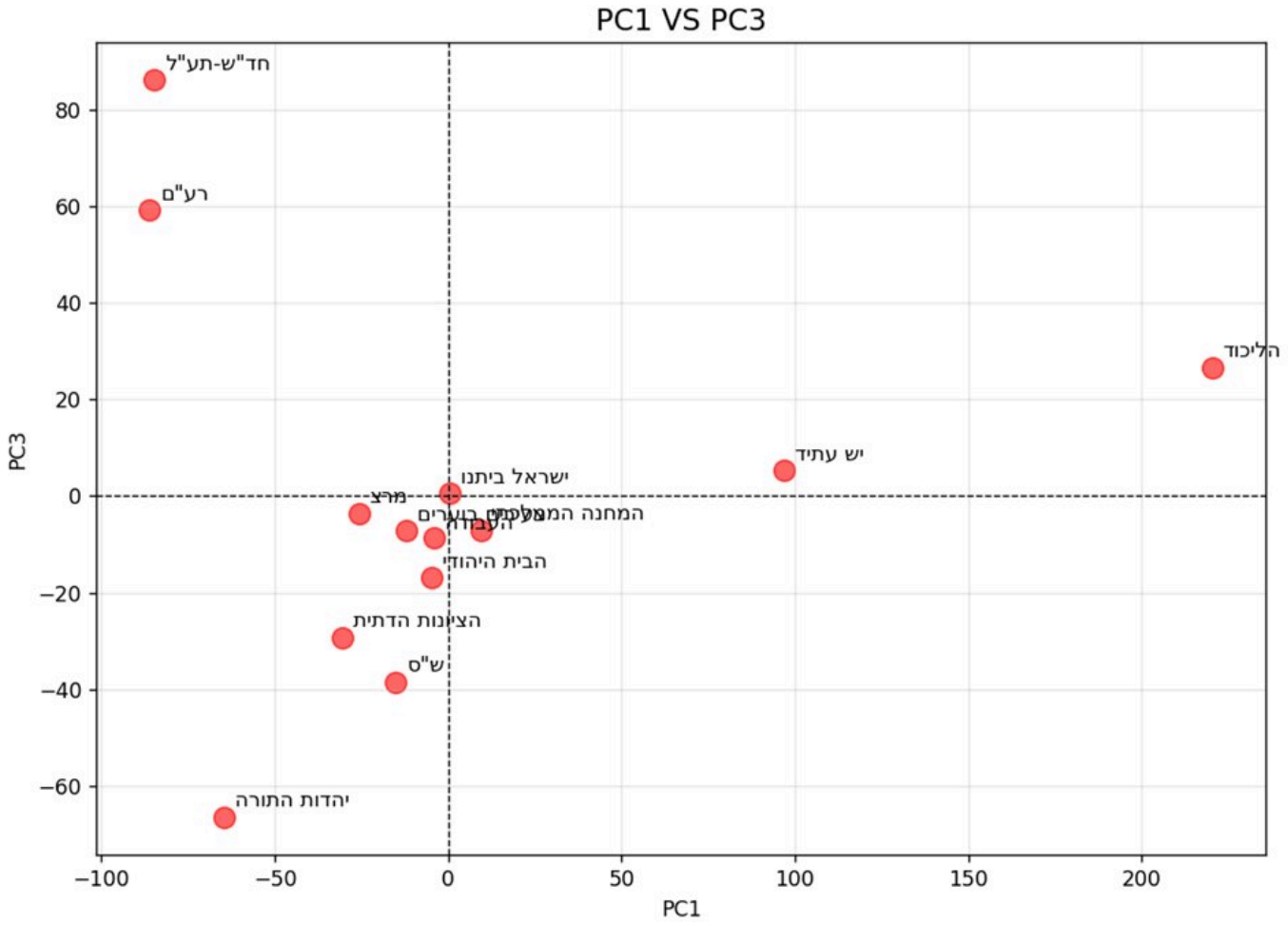
The distance between points shows that voting patterns in Israel are determined simultaneously by sectoral identity and by political-social position.

Figure labels

הליכוד	Likud	יש עתיד	Yesh Atid
הציונות הדתית	Religious Zionism	המחנה הממלכתי	National Unity
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	חד"ש-תע"ל	Hadash-Ta'al
הרשימה הערבית המאוחדת	United Arab List (Ra'am)	בל"ד	Balad
הבית היהודי	Jewish Home	צעירים בוערים	Tze'irim Bo'arim

Q3(C)

Ballot-box PCA: PC1 and PC3



This chart reveals the relationship between the national divide and the split within the right-wing bloc.

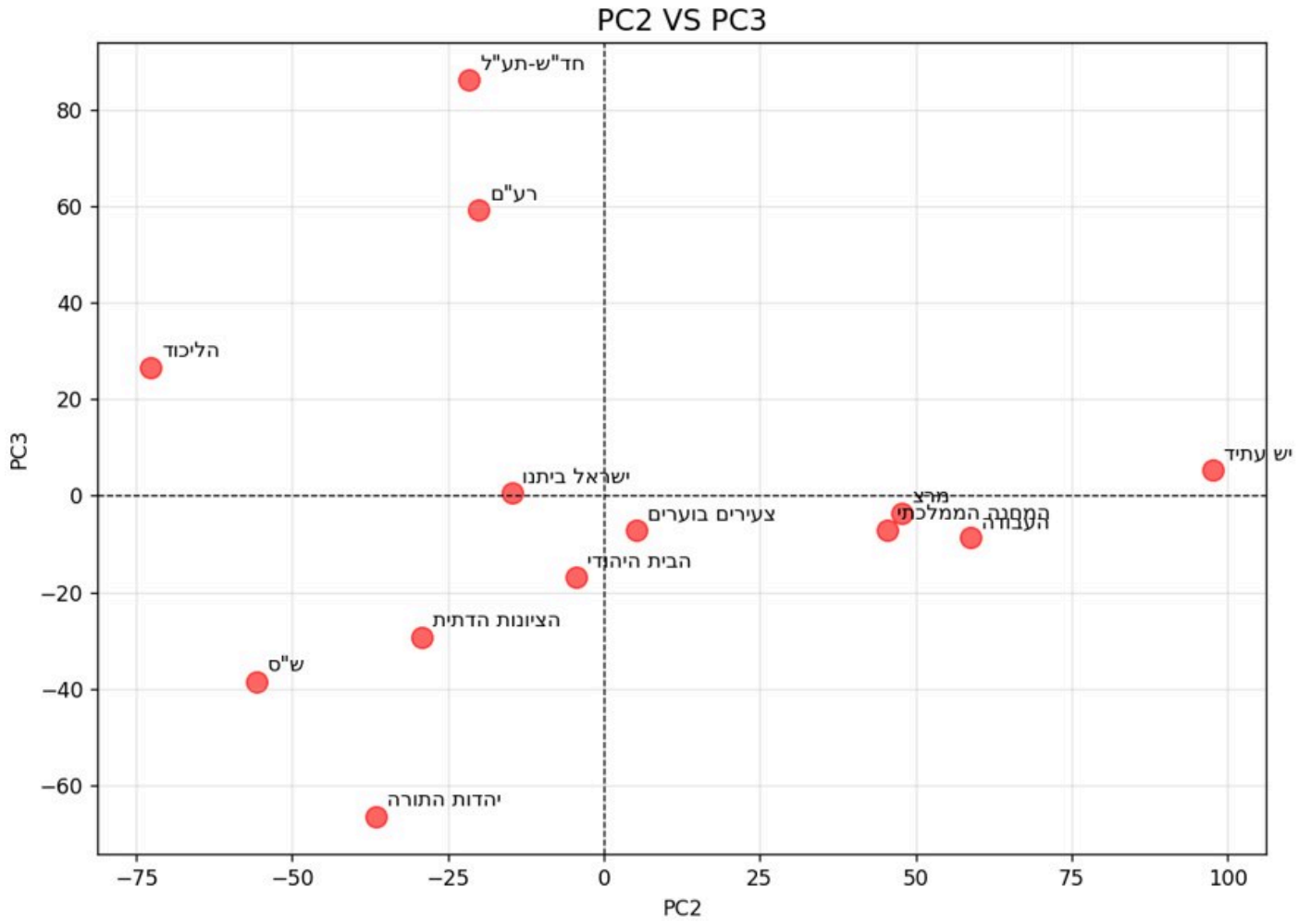
While PC1 separates the center and left from the extremes, PC3 creates a clear vertical separation within the religious camp: the Haredi parties (Shas and United Torah Judaism) move away from Likud and Religious Zionism. In this dimension, religious-community identity becomes the main distinguishing factor.

Figure labels

הליכוד	Likud	יש עתיד	Yesh Atid
הציונות הדתית	Religious Zionism	המחנה הממלכתי	National Unity
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	חד"ש-תע"ל	Hadash-Ta'al
הרשימה הערבית המאוחדת	United Arab List (Ra'am)	בל"ד	Balad
הבית היהודי	Jewish Home	צעירים בוערים	Tze'irim Bo'arim

Q3(C)

Ballot-box PCA: PC2 and PC3



This chart shows where the general ideological divide meets sectoral-religious identity.

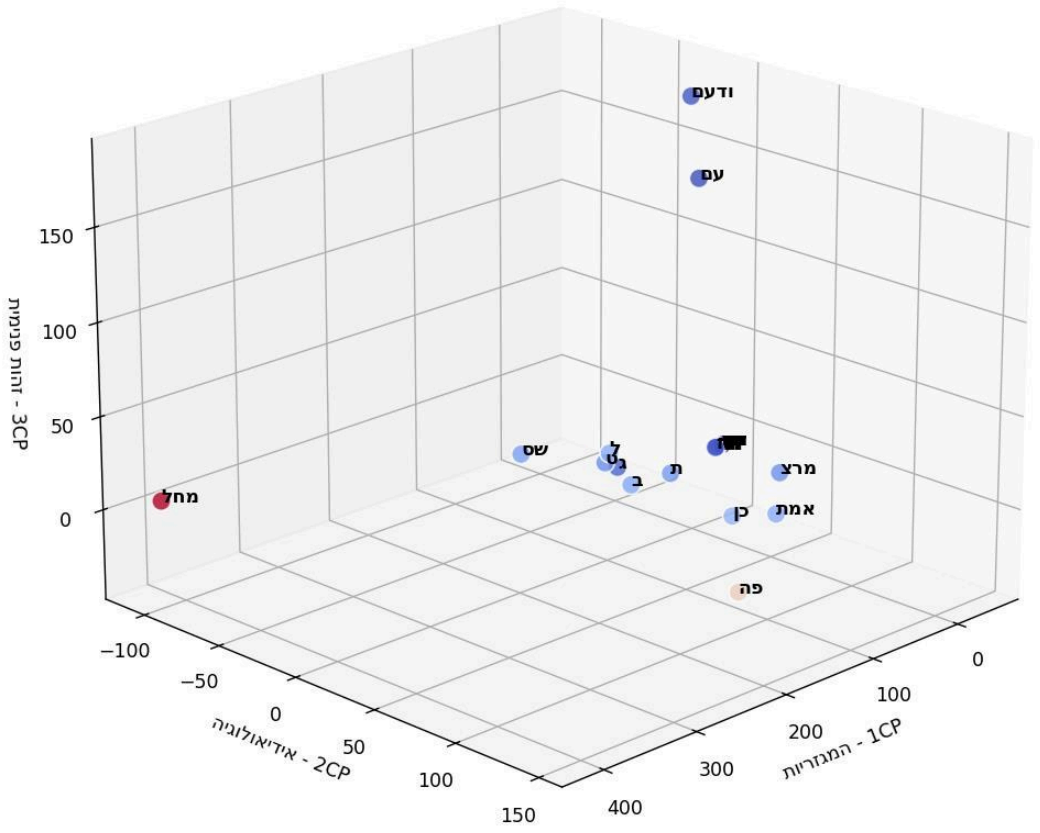
PC2 groups the whole bloc (Likud, Religious Zionism and the Haredi parties) against the left and Arab parties, while PC3 splits the bloc from within. Even within a unified political camp there is deep statistical variance arising from lifestyle and community identity.

Figure labels

הליכוד	Likud	יש עתיד	Yesh Atid
הציונות הדתית	Religious Zionism	המחנה הממלכתי	National Unity
ש"ס	Shas	יהדות התורה	United Torah Judaism
ישראל ביתנו	Yisrael Beiteinu	העבודה	Labor
מרצ	Meretz	חד"ש-תע"ל	Hadash-Ta'al
הרשימה הערבית המאוחדת	United Arab List (Ra'am)	בל"ד	Balad
הבית היהודי	Jewish Home	צעירים בוערים	Tze'irim Bo'arim

Q3(C)

Ballot-box model summary



- **PC1, the national divide:** separates the liberal center (positive pole) from the Arab sector (negative pole).
- **PC2, the ideological divide:** separates the religious-right bloc (negative pole) from the left-center and Arab bloc (positive pole).
- **PC3, intra-sector identity:** distinguishes within the right-wing bloc between the Haredi sector (negative pole) and the traditional national right (positive pole).

The axis labels in this figure are party codes and component names in Hebrew.

Q3(C)

Can 2024 and 2025 ballot-box results be compared?

Yes, at a qualitative level, but direct numerical comparison of the values requires care.

The mean moves

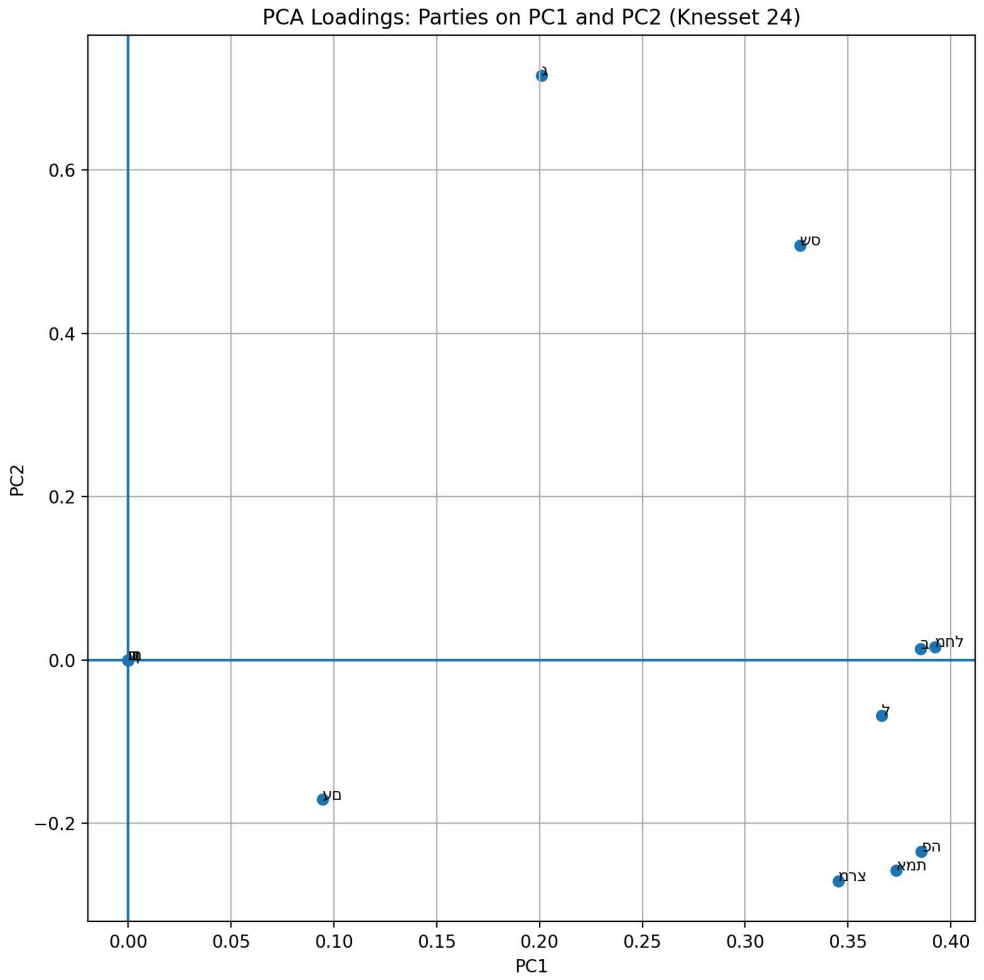
PCA is based on distances from the national mean (Z-score). If the whole country shifted in 2025, for example slightly to the right, the new mean would absorb that movement. A party can appear in the same position even though it actually grew stronger, simply because everyone grew with it.

The axes are redefined

Each year PCA searches again for the direction of greatest variance. If in 2025 the national divide became more dominant than the religious divide, PC1 of 2025 will look different from PC1 of 2024. It is like comparing two maps when the scale of one is stretched.

Q4 I–II

Component loadings, Knesset 24



Matrix size	Rows	Columns
2 × 13	PC1, PC2	13 parties

PC1: main political axis

Distinguishes parties with different ideological foundations; parties from the same political bloc receive similar values. Proximity on this axis matches the earlier similarity analysis, so PC1 can be interpreted as a main ideological division between blocs, such as right and left or coalition and opposition.

PC2: sectoral axis, secondary voting patterns

Explains further variance not explained by the main ideological axis. Sectoral parties, such as Haredi parties or those addressing distinct populations, differ from nationwide parties along it. PC2 can be interpreted as sectoral voting patterns or demographic characteristics that do not necessarily overlap with the main political division.

Points are labeled with party codes.

Q4 III

Why use one shared coordinate system?

Against separate axes for each election

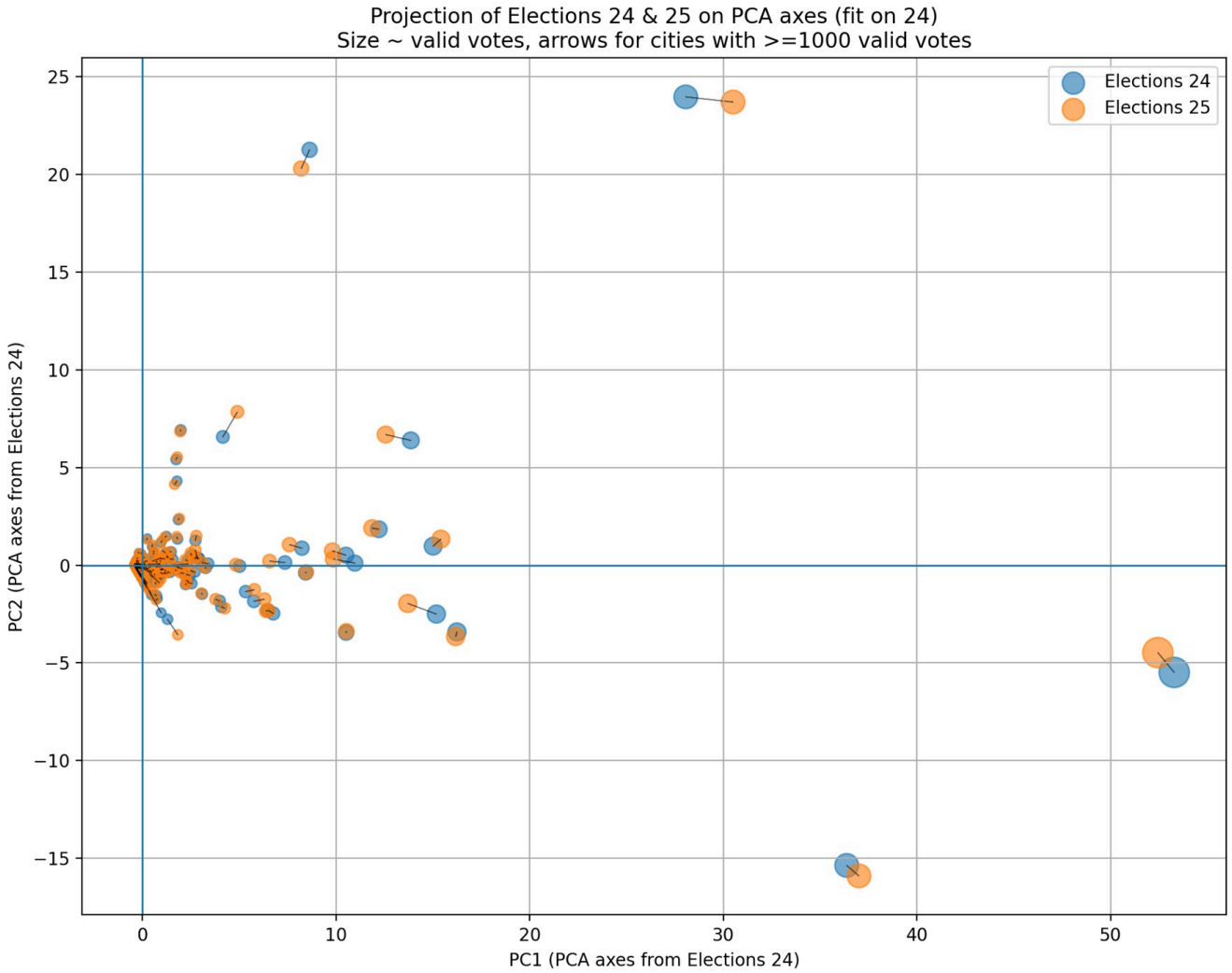
Projecting both elections onto one coordinate system gives a uniform frame of reference that allows direct comparison. Running PCA separately for each dataset produces different axes that are not comparable between years, so no consistent conclusions can be drawn. A single system ensures that observed differences come from changes in the data rather than in the definition of the axes.

Against one PCA on combined data

Fitting PCA on Knesset 24 only and projecting Knesset 25 onto those axes gives a fixed reference for examining change. Combining both datasets produces axes influenced by the whole period, which does not isolate the change relative to one fixed year. Using 2024 as the basis prevents bias in the axes and allows clear interpretation of the change.

Q4 IV

Projection of both elections on Knesset 24 axes



Q4 V

Conclusions: change between the elections

A change in voting patterns between Knesset 24 and Knesset 25 can be identified, and the figure clearly supports this.

Along PC1

Interpreted as the main ideological axis, most localities show only moderate changes. Arrows are usually short and point in a similar direction, indicating that the general ideological affiliation of localities remained relatively stable.

Along PC2

Interpreted as the axis separating sectoral patterns from broad appeal, more significant shifts appear in several localities, particularly large ones (marked with arrows). These indicate changes in voter composition, not necessarily a sharp move between blocs.

Bottom line: voting patterns underwent moderate change, mostly along the second axis representing sectoral characteristics, while the main ideological axis remained relatively stable. The changes do not reflect a broad ideological reversal, but adjustments within the same political space, such as strengthening or weakening of sectoral parties or shifts of votes between close parties.